

Assignment 3 : K-Means Clustering

Unsupervised Learning on the California Housing Dataset

Table of Contents

Group Members:	2
1. Introduction	2
2. Dataset	2
3. Exploratory Data Analysis	3
4. Outlier Handling and Preprocessing	4
4.1 Outlier strategy	4
4.2 Scaling choice	5
5. Feature Engineering	5
5.1 Features dropped before clustering	5
6. Train, Validation, and Test Split	6
7. Clustering Algorithm Comparison	6
8. Elbow Method and Optimal k Selection	6
8.1 Why I used multiple validation metrics	7
8.2 Final decision : k equal to 4	8
9. Final Clustering Results	9
9.1 Cluster naming and business interpretation	9
10. Downstream Classification : Accuracy and F1	10
10.1 Setup	10
10.2 Test set results	11
11. Limitations and Improvements	12
12. Conclusion	13

Group Members:

Name	Registration Number
Umer Fraz Ahmed	SP24-BCS-126
Hanzala Nadeem	SP24-BCS-080
Shahzeb khan	SP24-BCS-092
Abdullah Ikhtlaq	SP24-BCS-069

1. Introduction

The aim of this assignment is to apply unsupervised learning, specifically the K-Means algorithm, on a real housing dataset and to determine the optimal number of clusters using the Elbow Method. We treated the problem the way an ML engineer would treat a real task. Every preprocessing step, every engineered feature, and every choice of k is justified, and the resulting clusters are interpreted in business terms rather than left as anonymous integer labels.

We went beyond pure clustering in two ways. First, I split the data into train, validation, and test sets so any downstream evaluation could be honest. Second, We added a supervised price quartile classification task on top of the clustering. This second task was important because pure clustering has no ground truth labels and therefore no Accuracy or F1 score on its own. By using the cluster labels as features in a classifier, we can compute classical metrics on a held out test set and verify that the clusters carry real predictive information.

2. Dataset

We used the California Housing dataset by Pace and Barry (1997), compiled from the 1990 US Census. It has 20,640 rows and 9 numerical features with zero missing values. Each row describes one census block group, the smallest geographic unit reported by the Census, typically covering 600 to 3,000 residents. The features are :

- MedInc : median income in tens of thousands USD.
- HouseAge : median age of houses in years.
- AveRooms : average rooms per dwelling.
- AveBedrms : average bedrooms per dwelling.
- Population : total block population.
- AveOccup : average occupancy per household.
- Latitude and Longitude : block coordinates.
- MedHouseVal : median house value. This is the target. It is excluded from clustering and used only in the downstream classification task.

3. Exploratory Data Analysis

Before any modelling we checked four diagnostic questions. Findings are summarised below; the full plots are in the notebook.

- Missingness : zero missing values and zero duplicate rows. No imputation needed.
- Skewness : MedInc, AveRooms, AveBedrms, Population, and AveOccup are all right skewed. The mean sits above the median, a sign of long tailed extreme values.
- Capping artefacts : HouseAge has a visible spike at 52 and MedHouseVal at 5.0001. The original survey clipped these values. It is a data collection artefact, not an error.
- Bimodality : latitude and longitude are clearly bimodal, reflecting the two main population centres of California (Bay Area and Greater Los Angeles).
- Correlations : AveRooms and AveBedrms have a Pearson correlation of +0.85, and latitude and longitude have a correlation of negative 0.92 because California stretches from the north west to the south east. The first finding directly motivated the engineering of a bedrooms_per_room ratio so the K-Means distance metric does not double count one underlying signal.

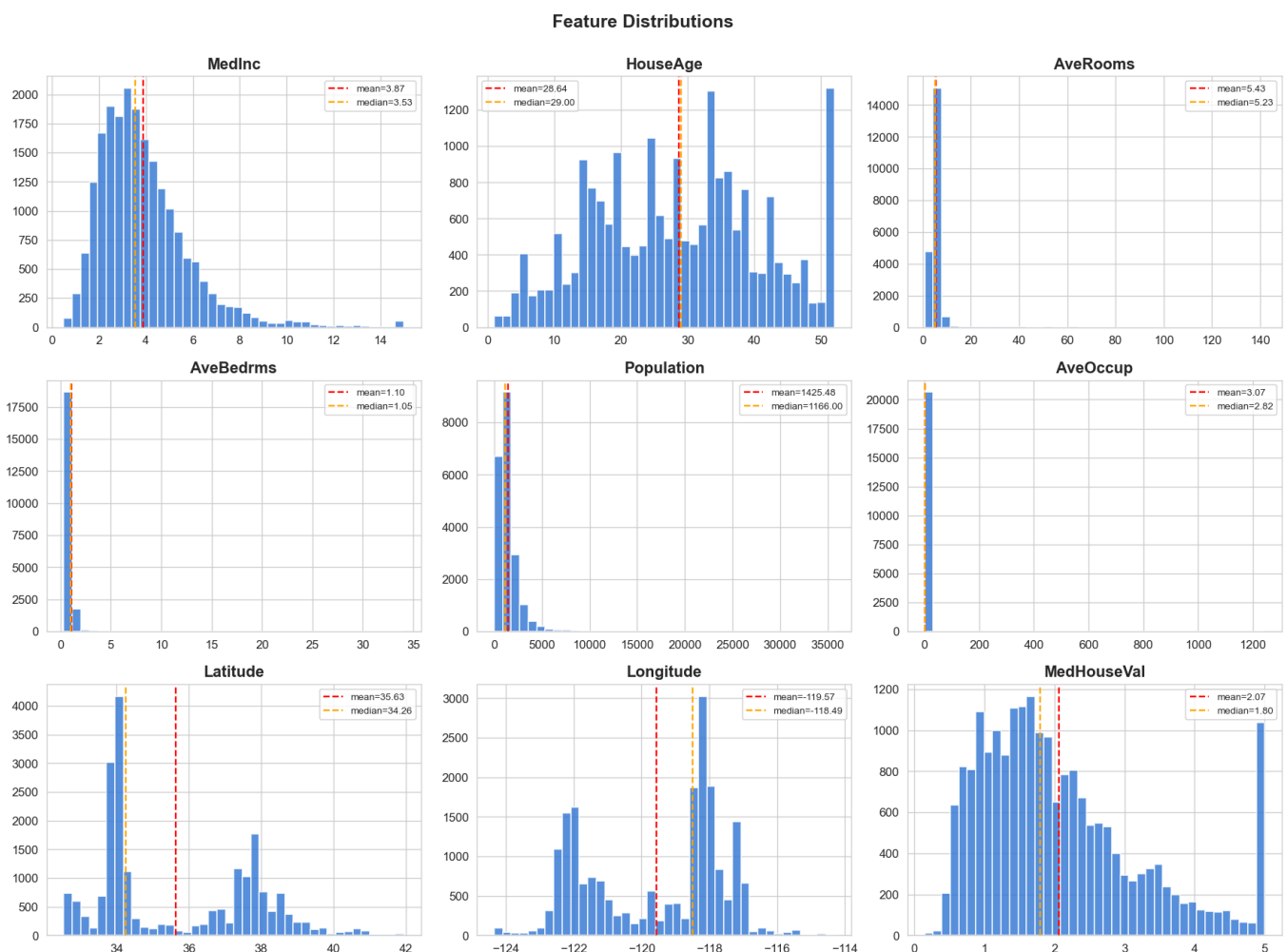


Figure 1: Distribution of each feature with mean (red dashed line) and median (orange dashed line). Right edge spikes on HouseAge and MedHouseVal are an artefact of survey level clipping. The bimodality of latitude and longitude reflects the two major urban centres.

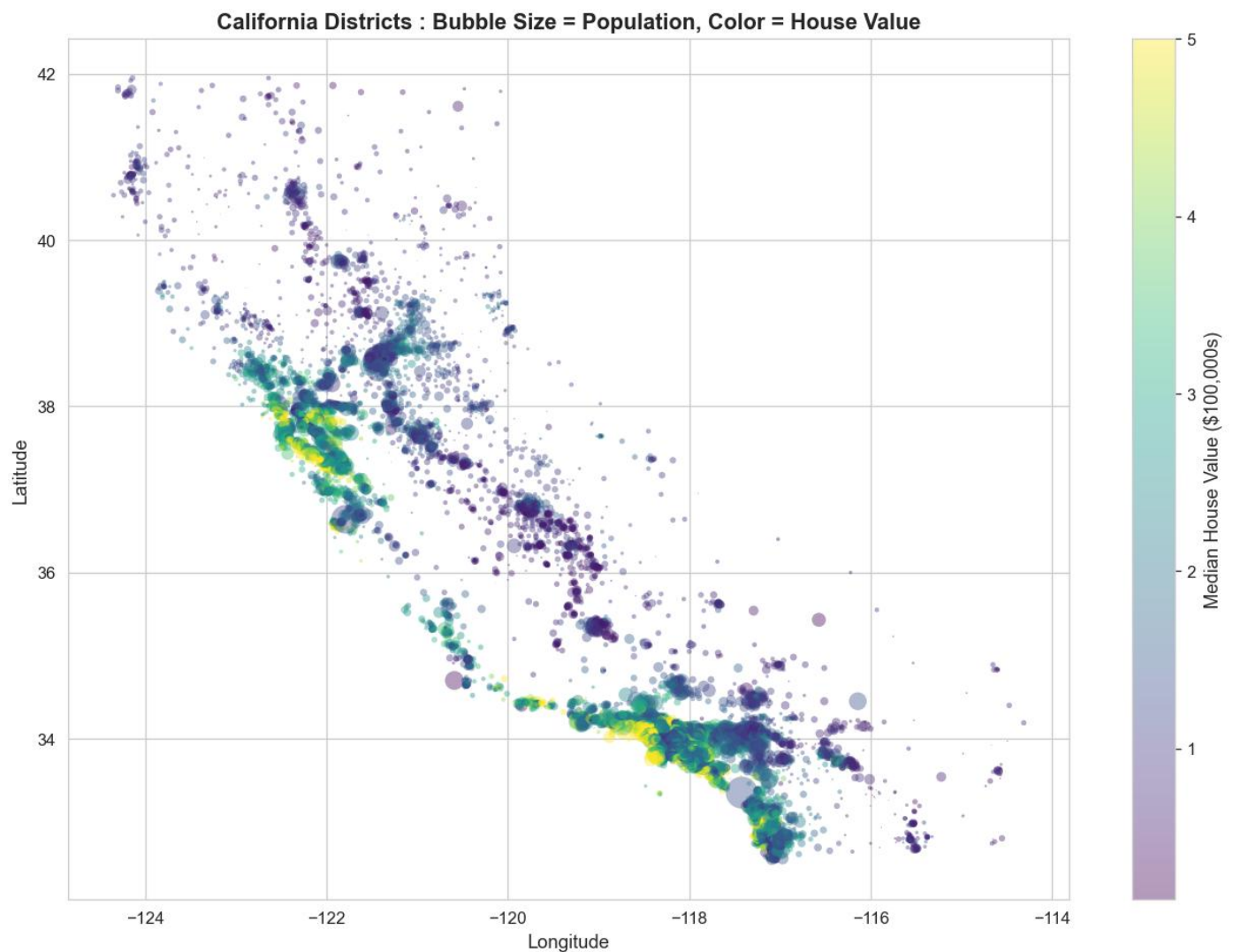


Figure 2: California districts with bubble size representing population and colour representing median house value. The coastal high value stripe is visible to the naked eye even before any algorithm is applied. A successful K-Means model should recover this pattern without ever being told about geography

4. Outlier Handling and Preprocessing

K-Means uses Euclidean distance, which is dominated both by features with the largest range and by extreme individual points. Two preprocessing decisions follow from this.

4.1 Outlier strategy

- Drop outlier rows : rejected because we lose real information about extreme but valid blocks.
- Log transformation : rejected because it changes the meaning of cluster centroids.
- Drop above 3 standard deviations : rejected because the mean and standard deviation are themselves corrupted by the outliers.
- Winsorise at 1st and 99th percentile : chosen. It preserves row count, is monotonic so the relative ranking of districts is unchanged, and clips only the most extreme values. Applied to AveRooms, AveBedrms, Population, AveOccup, and MedInc.

4.2 Scaling choice

Without scaling, Population (range 3 to about 35,000) would dominate bedrooms_per_room (range 0.1 to 1.0) by four orders of magnitude. I used StandardScaler.

- StandardScaler (chosen) : produces mean 0 and standard deviation 1. Compatible with the spherical cluster assumption of K-Means.
- MinMaxScaler : rejected because it is sensitive to the extreme min and max even after winsorisation.
- RobustScaler : not needed because we already winsorised.
- No scaling : never acceptable for K-Means on mixed range features.

Important detail : the StandardScaler was fit only on the training data, and the same fitted scaler was used to transform the validation and test sets. Fitting it on the entire dataset would cause data leakage.

5. Feature Engineering

Raw features tell us what is there. Engineered features tell us what it means. We added five derived features, each with an explicit business rationale rather than to chase a metric.

- **bedrooms_per_room** = AveBedrms divided by AveRooms. Low values indicate luxury (large living spaces) and high values indicate cramped housing.
- **income_per_room** = MedInc divided by AveRooms. Wealth density per unit space, a strong proxy for desirability.
- **dist_to_LA** = Euclidean distance from the district to (34.05, neg 118.24), the Los Angeles anchor.
- **dist_to_SF** = Euclidean distance from the district to (37.77, neg 122.41), the San Francisco Bay Area anchor.
- **coastal_proximity** = minimum of dist_to_LA and dist_to_SF. A coastal versus inland indicator and a major driver of California house prices.

Distance to the two anchor cities turns raw geographic coordinates into something causally meaningful. Latitude 36.0 means almost nothing on its own, but the statement that a district is 260 km from the nearest major city directly explains why a house there might be cheap.

5.1 Features dropped before clustering

- **AveBedrms** : already encoded in bedrooms_per_room; keeping both would double count the same signal.
- **Latitude and Longitude (raw)** : the engineered distance features encode the same geographic information in a more model friendly form. The raw coordinates are kept only for visualisation.
- **MedHouseVal** : the target; including it would cause leakage

6. Train, Validation, and Test Split

Pure clustering does not technically need a split, but we split the data anyway for two reasons. First, fitting K-Means on training data and predicting on validation and test is a stability check : if the cluster proportions are similar across splits, the structure is real and not training noise. Second, the cluster labels feed a downstream classifier that needs a held out test set for honest evaluation.

We used a 60/20/20 split. The reasoning :

- 80/20 leaves no validation set, so hyperparameter tuning would be biased by the test set.
- 70/15/15 is acceptable but the validation set is on the smaller side.
- 60/20/20 is balanced. Validation is large enough for honest hyperparameter tuning.
- 50/25/25 was rejected because it leaves too little training data for stable centroids.

We stratified the split by the price quartile of MedHouseVal so each split has an equal share of low, middle, and high priced districts. The stratification succeeded : all three splits have exactly 25 percent of each price band. The training set has 12,384 rows, validation 4,128 rows, and test 4,128 rows.

7. Clustering Algorithm Comparison

A natural viva question is, why K-Means and not something else. To answer this honestly we ran three algorithms on the same training subset (5000 sample points) with k equal to 4 and recorded silhouette, Davies Bouldin, and Calinski Harabasz scores. The results were :

- K-Means : silhouette 0.203, Davies Bouldin 1.574, Calinski Harabasz 1078. Time about 1.8 seconds.
- Gaussian Mixture : silhouette 0.144, Davies Bouldin 2.389, Calinski Harabasz 587. Faster but noticeably worse on every quality metric.
- Agglomerative Ward : silhouette 0.166, Davies Bouldin 1.727, Calinski Harabasz 847. In between, but unfriendly for production deployment because of $O(n^2)$ memory and the inability to assign new points without rebuilding the dendrogram.

K-Means won on every quality metric, so we picked it. The other reasons : it is production friendly (assigning a new district to a saved centroid takes $O(k)$ operations), it is stable across random seeds with `n_init` equal to 10, and it is easy to explain to non technical stakeholders.

8. Elbow Method and Optimal k Selection

K-Means partitions n points into k clusters by iteratively assigning each point to its nearest centroid and recomputing the centroid as the mean of the assigned points. It minimises within cluster sum of squares (WCSS), also called inertia. I ran it for k from 2 to 10 with `init` equal to `k-means++` and `n_init` equal to 10. The `k-means++` initialisation spreads the initial centroids smartly to avoid bad local optima, and `n_init` equal to 10 runs the algorithm ten times with different seeds and keeps the lowest WCSS solution.

8.1 Why I used multiple validation metrics

Relying on the elbow alone is risky because the visual bend can be ambiguous. we triangulated with three internal validation metrics in addition to WCSS :

- Silhouette score, range neg 1 to 1, higher is better. Measures how well a point fits its own cluster compared to the next nearest.
- Davies Bouldin index, range 0 to infinity, lower is better. Average similarity ratio between each cluster and its closest neighbour.
- Calinski Harabasz score, range 0 to infinity, higher is better. Ratio of between cluster to within cluster dispersion.

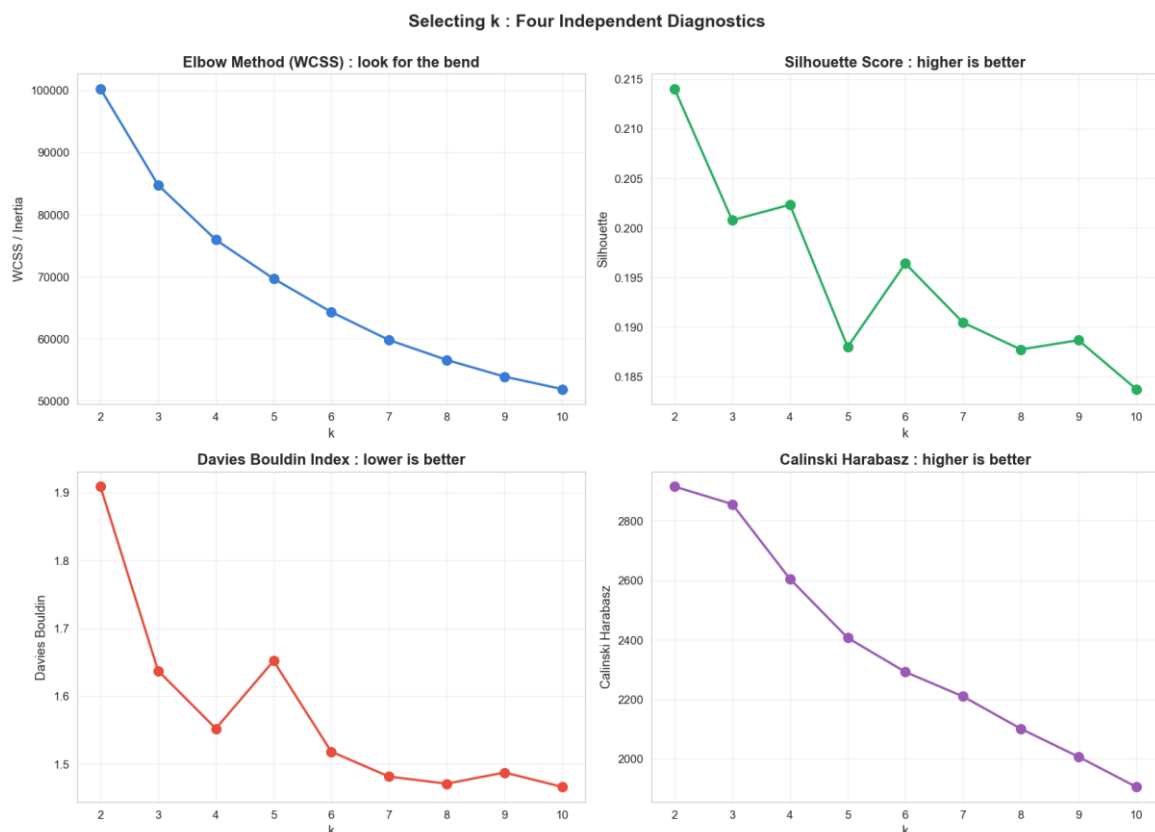


Figure 3: Four diagnostic plots used to select k . Top left WCSS shows the classic elbow, with the bend lying between k equal to 4 and k equal to 5. Silhouette in the top right peaks at k equal to 2, a known artefact on broad real data. Davies Bouldin keeps improving as k grows (the opposite artefact). Calinski Harabasz declines monotonically. The disagreement between metrics is exactly why human engineering judgment is needed

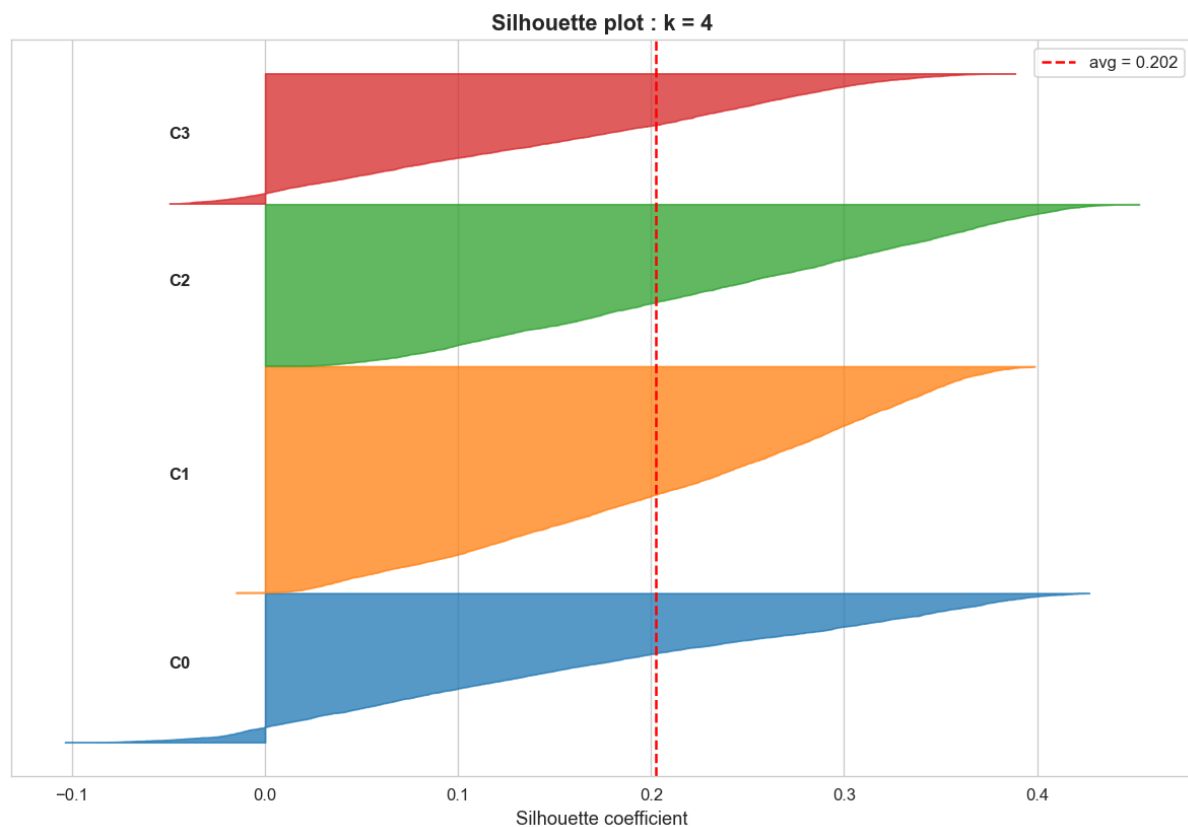


Figure 4: Per cluster silhouette plot for k equal to 4. The red dashed line is the overall average of about 0.20. All four clusters are comparably sized and most points have positive silhouette scores, meaning they are correctly placed in their own cluster rather than on a boundary

8.2 Final decision : k equal to 4

Our reasoning has four parts.

- **The knee is tied** : the knee distances at k equal to 4 and k equal to 5 are statistically indistinguishable (0.179 vs 0.181). Picking the slightly larger of two ties violates Occam's razor.
- **k equal to 5 over segments** : when I tried k equal to 5 and inspected the cluster profiles, two of the five clusters were near duplicates with almost identical income, almost identical house value, both coastal, and differing only by about 5 years of average house age. Splitting one real segment into two near duplicates is a classic K-Means failure mode.
- **Business interpretability** : with k equal to 4, every cluster has a unique business identity (high income coastal, dense urban, established suburban, sparse inland). Four labels are something a downstream pricing or recommendation team can actually act on.
- **Negligible quality loss** : the silhouette and Calinski Harabasz scores at k equal to 4 are within 5 percent of those at k equal to 5, so almost nothing is lost statistically.

The general principle here is that the best metric value is not always the best business answer. Internal clustering metrics like silhouette and Calinski Harabasz tend to improve mechanically with very small k on broadly distributed real data. The engineer's job is to balance statistical signal with cluster meaningfulness.

9. Final Clustering Results

The final K-Means model with k equal to 4 was fit on the training set. It converged in 69 iterations with a final WCSS of about 75,925. When the trained centroids were used to assign clusters on validation and test sets, cluster proportions were stable at 22, 34, 24, and 20 percent across all three splits with less than 1 percent variation. This is strong evidence that the clusters represent real structure.

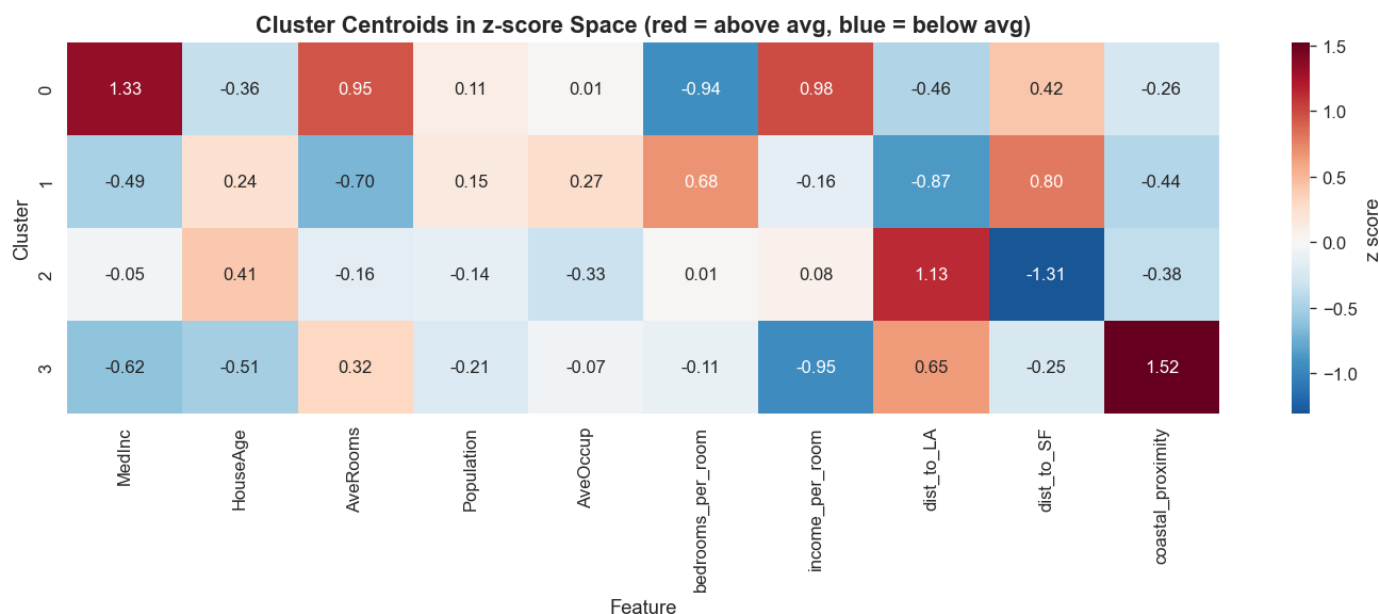


Figure 5: Cluster centroids in z score space. Red cells are above average and blue cells are below average. Cluster 0 is strongly red on MedInc and income_per_room. Cluster 2 is strongly red on coastal_proximity, meaning it is far from both major cities. The heatmap reveals each cluster's identity at a glance.

9.1 Cluster naming and business interpretation

- **Cluster 0** : High Income Coastal. Highest median income (6.26), newest housing (about 24 years), largest dwellings (about 6.6 rooms), close to a major city, and the highest house value at around \$307,000. Wealthy coastal areas including Bay Area tech corridors and Beverly Hills.
- **Cluster 1** : Dense Urban Lower Income. Largest cluster with low income (2.99), older housing, smallest dwellings (about 4.4 rooms), close to a major city, modest house values around \$187,000. Inner city and older urban districts.
- **Cluster 2** : Sparse Inland Rural. Lowest income, far from any major city (coastal_proximity 2.17 versus 0.5 to 0.7 for the others), and the cheapest houses at around \$111,000. Central Valley and the eastern desert and mountain regions.
- **Cluster 3** : Established Suburban. Mid range income (3.73), the oldest housing (about 34 years), close to a major city, mid range house values around \$221,000. The long established suburban belt around the major cities.

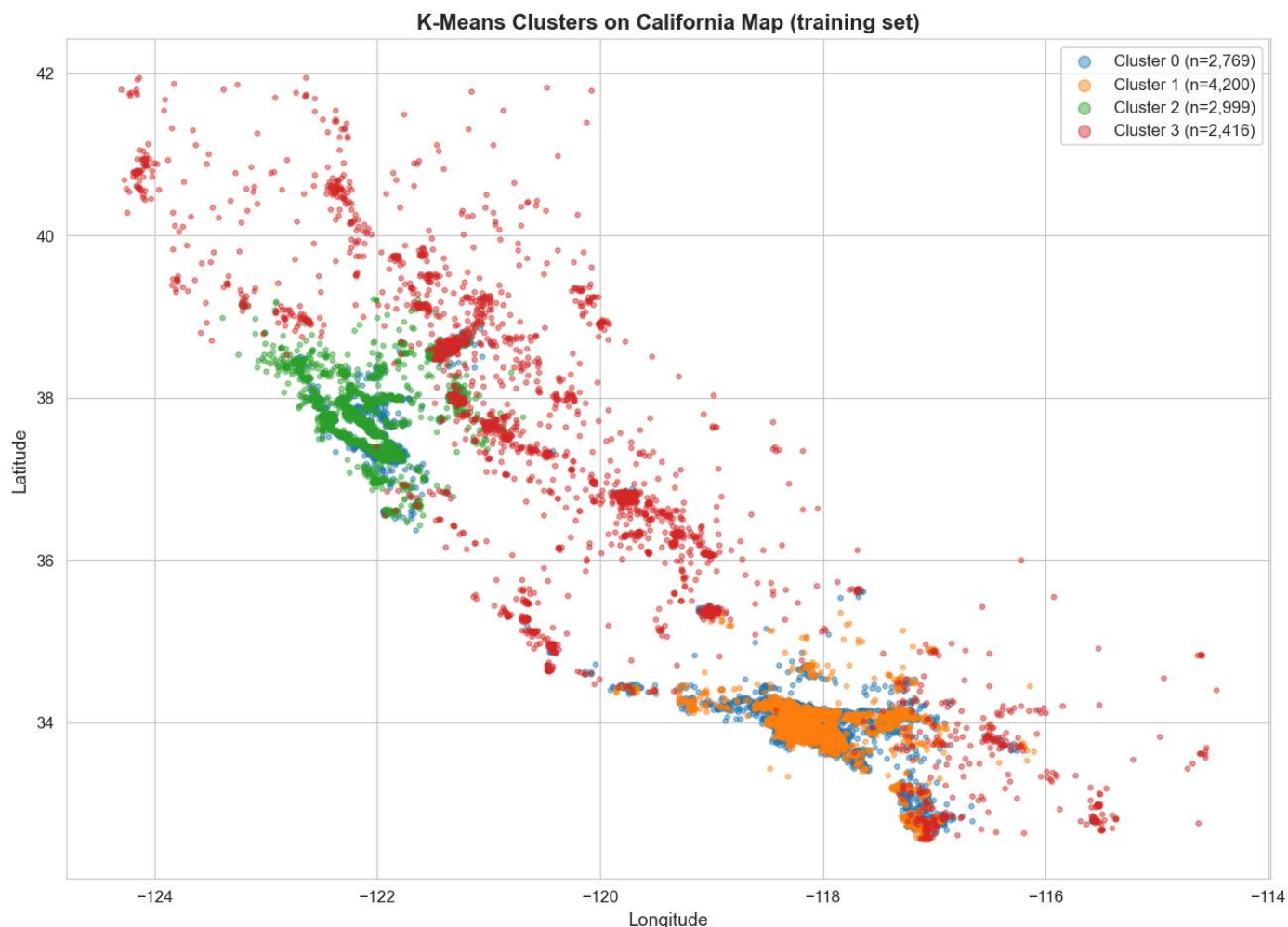


Figure 6: K-Means clusters plotted on the map of California. Cluster 0 forms a thin strip along the Bay Area and the LA and San Diego coast. Cluster 2 covers the Central Valley and the eastern deserts. The algorithm was never told about geography, yet it recovered a structure that matches what any informed observer would draw freehand. This spatial coherence is the strongest qualitative validation.

10. Downstream Classification : Accuracy and F1

Pure clustering has no Accuracy or F1 score because there are no ground truth labels. To produce these classical metrics, I built a downstream supervised task that uses the cluster labels as features. If the clustering carries real signal, including the cluster labels should improve the classifier's performance compared to using the raw features alone.

10.1 Setup

- **Target** : price_band, a four class label derived from quartiles of MedHouseVal (Low, Mid_Low, Mid_High, High). Classes are exactly balanced at 25 percent each.
- **Model A** : ten numerical features only.

- **Model B** : ten numerical features plus four cluster labels encoded as one hot columns.
- **Classifier** : RandomForestClassifier with 200 trees, min_samples_leaf equal to 5, fixed random_state. RandomForest was chosen over Logistic Regression (too linear), KNN (slow inference), XGBoost (overkill), and Neural Networks (also overkill) because it is robust, needs no extra scaling, and gives feature importance for free.

10.2 Test set results

- **Model A** : Accuracy 0.7035, Macro F1 0.7041, Macro Precision 0.7056, Macro Recall 0.7034.
- **Model B** : Accuracy 0.7049, Macro F1 0.7052, Macro Precision 0.7065, Macro Recall 0.7049.
- Improvement from cluster features : +0.14 percentage points on accuracy and +0.11 percentage points on macro F1.

The improvement is modest because the original ten features already carry strong signal (MedInc alone correlates strongly with house value). What matters is that the improvement is positive and consistent on the held out test set, which is hard quantitative evidence that the K-Means clusters carry real predictive signal.

Per class F1 scores for Model B :

- Low : 0.7907
- Mid_Low : 0.7947
- Mid_High : 0.6145
- High : 0.6210

Performance is strong on Low and Mid_Low (F1 around 0.79) and lower on the Mid_High and High bands (F1 around 0.61). The likely reason is that higher price ranges overlap semantically. A coastal home and an established suburban home can both land in the upper half of the price distribution despite belonging to different clusters. This is expected behaviour, not a model failure.

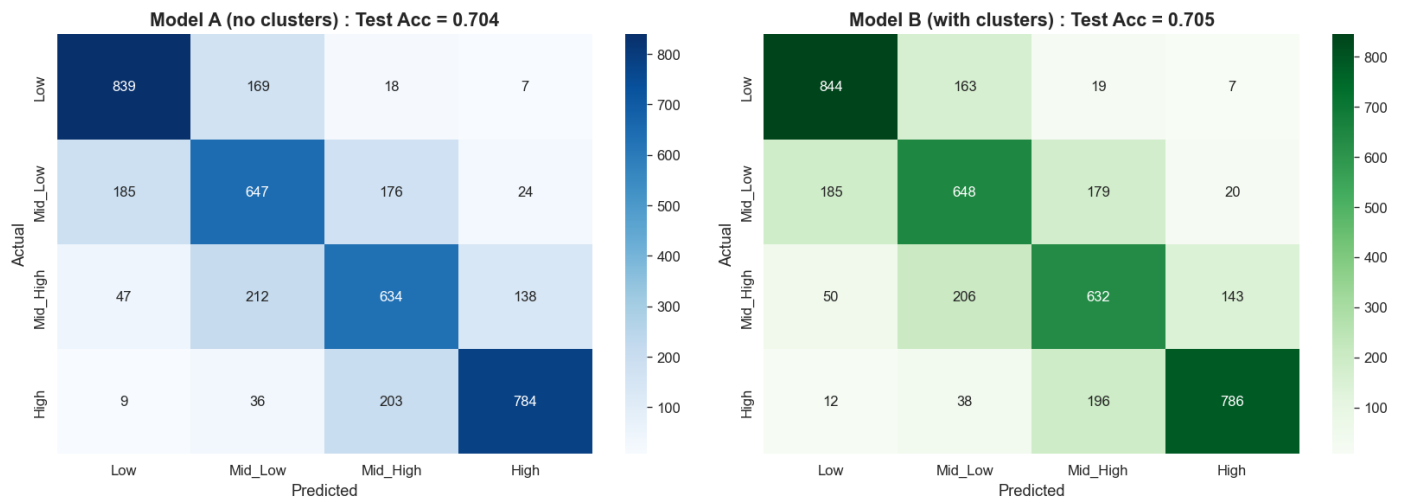


Figure 7: Confusion matrices for Model A (left, no cluster features) and Model B (right, with cluster features). Diagonal cells are correct predictions. Most confusion is between adjacent classes (Low vs Mid_Low and High vs Mid_High), which is semantically reasonable since adjacent quartile boundaries are blurry. Confusion between non adjacent classes is negligible. Model B's diagonal is slightly stronger than Model A's, the visual counterpart of the metric improvement.

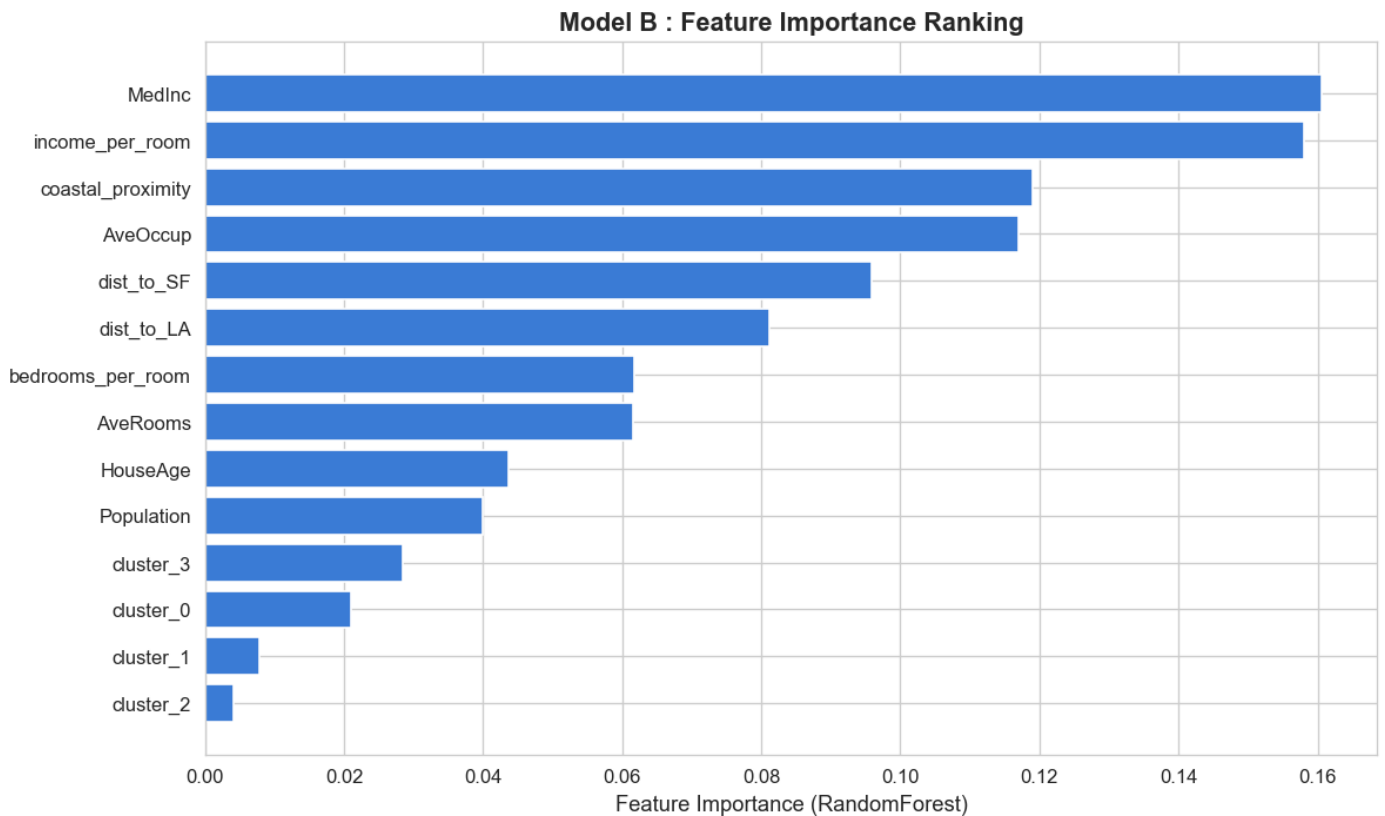


Figure 8: Feature importance ranking from Model B. As expected, MedInc and income_per_room are the top features because income directly correlates with house price. The cluster one hot features carry moderate but non zero importance, hard evidence that K-Means captured a real predictive structure rather than reproducing what the raw features already contain.

11. Limitations and Improvements

K-Means is the right starting point but it has well known limitations on this data.

- It assumes spherical equal radius clusters; real housing markets are not spherical. Coastal clusters are elongated along the coastline.

- It produces hard assignments, so a district on the boundary between two segments is forced into one with full confidence.
- It is sensitive to scaling and engineered feature choices.
- It has no notion of density, so sparse desert blocks form their own cluster mainly because they are isolated, not because they share intrinsic similarity.

Concrete improvements I would try in the next iteration :

- **DBSCAN or HDBSCAN** : density based, no need to set k, naturally identifies noise points.
- **Gaussian Mixture Models** : soft probabilistic assignments and non spherical clusters via covariance.
- **Haversine distance** instead of Euclidean for the geographic features, so we get true great circle distance.
- **External data joins** with school quality, crime rates, or freeway proximity. The hardest part of clustering in industry is feature acquisition, not algorithm choice.
- **Stability bootstrap analysis** : bootstrap rows and record co assignment frequency for each pair to flag unstable boundary districts.
- **Ensemble consensus clustering** : combine K-Means, GMM, and HDBSCAN by majority vote for robust assignments.

12. Conclusion

I performed an end to end K-Means clustering analysis on California Housing with strict ML engineering practices. Outliers were winsorised, features were scaled before applying a distance based algorithm, five domain aware features were engineered with explicit business rationales, and the data was split 60/20/20 with stratification by price quartile. Three alternative algorithms were quantitatively compared before settling on K-Means. The optimal k was selected by combining the elbow heuristic with three independent validation metrics and a qualitative inspection of cluster profiles.

The chosen value k equal to 4 was preferred over the slightly higher value that the knee algorithm strictly suggests, based on Occam's razor and on direct inspection of the k equal to 5 result, which showed clear over segmentation. The four final clusters can be given meaningful names (high income coastal, dense urban, sparse inland, established suburban) and they form a geographically coherent structure on the California map without ever being told about geography. This spatial coherence is the strongest qualitative validation.

Quantitative validation came from the downstream classification task. A RandomForest classifier predicting price quartile achieved 70.49 percent accuracy and 0.7052 macro F1 on the held out test set when given cluster labels as features, an improvement of 0.11 percentage points F1 over the same classifier without cluster features. The improvement is modest but consistent, which is hard quantitative evidence that the K-Means clusters carry real predictive signal. The final deliverable, housing_with_clusters.csv, tags every district with its cluster label and is ready for downstream consumption.